

PRACTICAL No. 0

Aim: Introduction of Cyber Security & Cryptography Lab.

Theory:

Cyber Security:

The Cyber Security & Privacy Lab is designed to help students understand the fundamental concepts, tools, and techniques used to protect digital information and systems. In today's world, where technology is deeply integrated into everyday life, securing data, networks, and communication channels has become essential. This lab provides a hands-on environment to learn and practice various security mechanisms that defend against cyber threats and ensure privacy.

1. How cyber attacks happen

You will understand how hackers try to break into systems, steal passwords, send viruses, or misuse networks.

2. How to protect systems

The lab teaches you different techniques and tools to secure computers and networks, such as:

- Password protection
- Firewall configuration
- Antivirus tools
- Network monitoring

3. Cryptography basics

Cryptography means protecting information using secret codes.

In the lab, you learn:

- How to encrypt (convert normal text into secret text)
- How to decrypt (convert secret text back to normal text)
- How algorithms like Playfair, DES, RSA, RC4, etc. work

4. Privacy protection

You learn how to keep personal and sensitive data safe, such as:

- User identity
- Banking information
- Personal files
- Communication details

5. Hands-on practical skills

The lab gives you real experience by performing tasks like:

- Encrypting and decrypting messages
- Implementing security algorithms
- Detecting threats

- Understanding cyber laws
- Learning ethical hacking basics

Cryptography:

Cryptography is the science to encrypt and decrypt data that enables the users to store sensitive information or transmit it across insecure networks so that it can be read only by the intended recipient.

Data which can be read and understood without any special measures is called plaintext, while the method of disguising plaintext in order to hide its substance is called encryption.

Encrypted plaintext is known as cipher text and process of reverting the encrypted data back to plain text is known as decryption.

- The science of analysing and breaking secure communication is known as cryptanalysis. The people who perform the same also known as attackers.
- Cryptography can be either strong or weak and the strength is measured by the time and resources it would require to recover the actual plaintext.
- Hence an appropriate decoding tool is required to decipher the strong encrypted messages.
- There are some cryptographic techniques available with which even a billion computers doing a billion checks a second, it is not possible to decipher the text.

A cryptographic algorithm works in combination with a key (can be a word, number, or phrase) to encrypt the plaintext and the same plaintext encrypts to different cipher text with different keys.

Hence, the encrypted data is completely dependent couple of parameters such as the strength of the cryptographic algorithm and the secrecy of the key.

Cryptography Techniques:

Symmetric Encryption – Conventional cryptography, also known as conventional encryption, is the technique in which only one key is used for both encryption and decryption. For example, DES, Triple DES algorithms, MARS by IBM, RC2, RC4, RC5, RC6.

Asymmetric Encryption – It is Public key cryptography that uses a pair of keys for encryption: a public key to encrypt data and a private key for decryption. Public key is published to the people while keeping the private key secret. For example, RSA, Digital Signature Algorithm (DSA), Elgamal.

Hashing – Hashing is ONE-WAY encryption, which creates a scrambled output that cannot be reversed or at least cannot be reversed easily. For example, MD5 algorithm. It is used to create Digital Certificates, Digital signatures, Storage of passwords, Verification of communications, etc.

OpenSSH/PuTTY/SSH

SSH (Secure Shell) is the now ubiquitous program for logging into or executing commands on a remote machine. It provides secure encrypted communications between two untrusted hosts over an insecure network, replacing the hideously insecure telnet/rlogin/rsh alternatives. Most UNIX users run the open source OpenSSH server and client. Windows users often prefer the free PuTTY client, which is also available for many mobile devices, and WinSCP. Other Windows users prefer the nice terminal-based port of OpenSSH that comes with Cygwin.

GnuPG

PGP is the famous encryption system originally written by Phil Zimmerman which helps secure your data from eavesdroppers and other risks. GnuPG is a very well-regarded open source implementation of the PGP standard (the actual executable is named `gpg`). While the excellent GnuPG is always free, PGP is now owned by Symantec and costs a lot of money.

OpenSSL

The OpenSSL Project is a collaborative effort to develop a robust, commercial-grade, full-featured, and open source toolkit implementing the Secure Sockets Layer (SSL v2/v3) and Transport Layer Security (TLS v1) protocols as well as a full-strength general-purpose cryptography library. Apart from being a component of many crypto programs, OpenSSL comes with a lot of command-line tools for encryption, hashing, certificate handling, and more.

Tor

Tor is a network of virtual tunnels designed to improve privacy and security on the Internet by routing your requests through a series of intermediate machines. It uses a normal proxy server interface so that ordinary Internet applications like web browsers and chat programs can be configured to use it. In addition to helping preserve users' anonymity, Tor can help evade firewall restrictions. Tor's hidden services allow users publish web sites and other services without revealing their identity or location. For a free cross-platform GUI, users recommend Vidalia. Remember that Tor exit nodes are sometimes run by malicious parties and can sniff your traffic, so avoid authenticating using insecure network protocols (such as non-SSL web sites and mail servers).

OpenVPN

OpenVPN is an open-source SSL VPN package which can accommodate a wide range of configurations, including remote access, site-to-site VPNs, WiFi security, and enterprise-scale remote access solutions with load balancing, failover, and fine-grained access-controls. OpenVPN implements OSI layer 2 or 3 secure network extension using the industry standard SSL/TLS protocol, supports flexible client authentication methods based on certificates, smart cards, and/or 2-factor authentication, and allows user or group-specific access control policies using firewall rules applied to the VPN virtual interface. OpenVPN uses OpenSSL as its primary cryptographic library.

Conclusion:

Thus we studied about the Cyber Security and Cryptography Lab.